

Cobalt-Catalyzed Cross-Coupling of Organozinc Halides with Bromoalkynes

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Abstract: The cobalt-catalyzed cross-coupling of organozinc bromides is described. In the case of arylzinc reagents, the same cobalt catalyst, $\text{CoBr}_2(\text{phen})$, can be used for both the synthesis of the organozinc species and the cross-coupling reaction. In the case of alkylzinc halides, which allow access to dialkylalkynes, the method used for the preparation of the organozinc species has a significant impact on the yields.

Keywords: alkynes; C–C coupling; cobalt; cross-coupling; zinc

The phenylacetylene motif is important in areas that range from organic synthesis^[1] to materials^[2] and polymer science.^[3] Usually, it is obtained through Pd-catalyzed cross-coupling of a terminal acetylene or metallic acetylide to a suitably functionalized arene, the best-known methodology being the Sonogashira cross-coupling.^[4] In such cases, the alkyne derivative is used as a nucleophile, and an aryl halide or pseudohalide is the electrophile. This elegant method has become the most important method for the formation of $\text{Csp}-\text{Csp}^2$ bonds. In this reaction, the use of hindered alkynes or of unactivated aromatic reagents (aryl chlorides, electron-rich aryl bromides) usually requires expensive ligands [$\text{P}(t\text{-Bu})_3$, PCy_3 ...]. Furthermore, this reaction is less efficient with aliphatic halides as electrophiles. The reverse case, where the alkyne moiety is used as the electrophilic partner, has received much less attention.^[5] The presence of a heteroatom substituent on the triple bond has a strong influence on the alkyne's reactivity. Thiomethylated alkynes have been coupled using palladium catalysts both to functionalized organozinc reagents (including arylzinc halides)^[6] and to arylboronic acids,^[7] in the same way as alkynyl iodonium salts under copper cat-

alysis.^[8] However, the access to halogenoacetylenes is usually easier but the sp hybridization of the triple bond confers a low reactivity to these products. Preliminary results shown that Grignard reagents reacted with haloalkynes under cobalt catalysis in low yields as reported by Kharash.^[9] Later, organomagnesium-derived arylcopper reagents have been coupled to bromoalkynes under iron catalysis.^[10] Equally, boronic acid derivatives have been coupled to haloalkynes under copper^[11] or palladium^[12] catalysis, but, it was only in 2010 that Cahiez's group reported a very general copper-catalyzed cross-coupling of Grignard reagents with alkynyl halides.^[13] Cross-coupling with alkylzinc compounds were reported more recently by Knochel using Pd catalysis.^[14] Despite these recent advances, the development of a mild, inexpensive and simple procedure remains highly desirable. As our group has already shown that arylzinc reagents can be formed very easily using a cobalt catalyst,^[15] and coupled with various electrophiles with the same catalyst,^[16] it is possible to envision a convenient methodology wherein a cobalt-catalyzed synthesis of an organozinc reagent is associated with its cobalt-catalyzed coupling to a haloalkyne.

This method potentially allows the formation of highly substituted phenylacetylene derivatives without requiring strong bases (e.g., $n\text{-BuLi}$) or expensive catalysts (e.g., Pd), enabling different strategies in the introduction of functional groups on this structural motif.

Having successfully developed methods using a CoBr_2/Zn system for the synthesis of arylzinc bromides, our preliminary study began with the reaction of p -methoxyphenylzinc bromide, formed using cobalt catalysis, with 1-bromophenylacetylene (Table 1). Adding the bromoalkyne directly to the crude solution of arylzinc bromide afforded the desired 4-methoxydiphenylacetylene in modest yield (Table 1, entry 1). As we have previously observed,^[17] the excess zinc still present in the reaction medium may

Table 1. Optimization of reaction conditions.

Entry	Ligand	Yield ^[b]
1	none ^[c]	45%
2	none	57%
3	pyridine	78%
4	2,2'-bipyridine (13 mol%)	55%
5	1,10-phenanthroline ^[d]	79% ^[e]
6	PPh₃ (26 mol%)	89% ^[e]
7	PPh ₃ (26 mol%) ^[f]	42%
8	DPPE (13 mol%)	70%

^[a] ArZnBr was titrated by GC on iodolyzed aliquots, with decane as internal standard.

^[b] GC yield (vs. bromoalkyne, decane as internal standard).

^[c] ArZnBr was not filtered.

^[d] CoBr₂(phen) (10 mol%) was used to prepare ArZnBr.

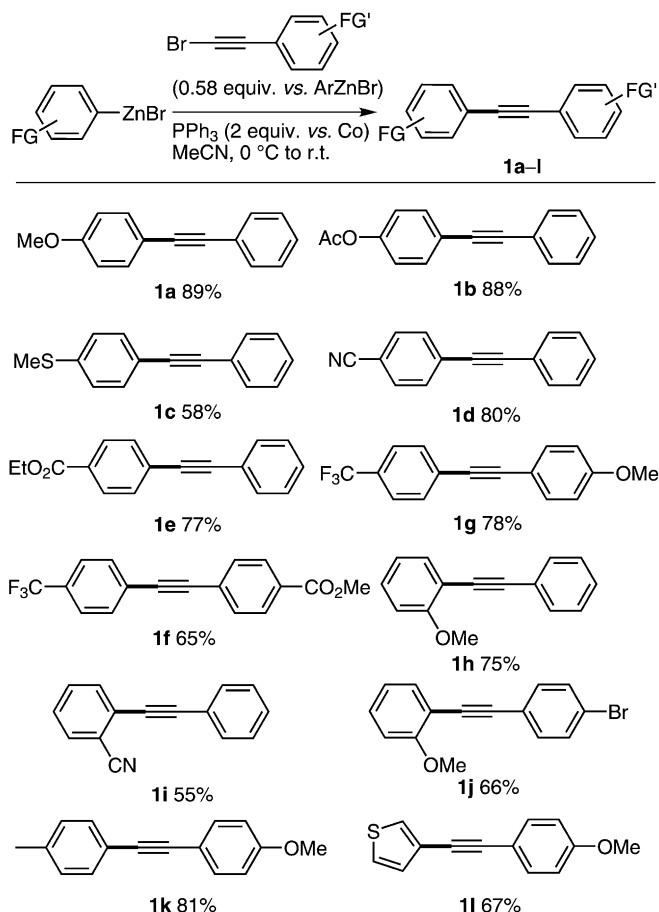
^[e] Isolated yield.

^[f] Bromoalkyne added at room temperature.

be problematic with alkynes. However, it can easily be removed by filtration (see ref.^[17] and the Experimental Section) and this allows the yield to be raised to 57% (Table 1, entry 2). The addition of ligands was examined, triphenylphosphine proving to be the most effective (Table 1, entries 3 to 8). Introduction of the bromoalkyne at 0 °C was also crucial to obtain good yields (Table 1, entries 6 and 7). Reactions using these conditions, with a smaller excess of arylzinc bromide resulted in complex mixtures.

Next, we set out to explore the scope of the methodology. A variety of arylzinc bromides bearing electron-donating or electron-withdrawing groups was coupled with 1-bromophenylacetylene derivatives under the optimized conditions above to afford functionalized diphenylacetylenes in moderate to excellent yield (Scheme 1).

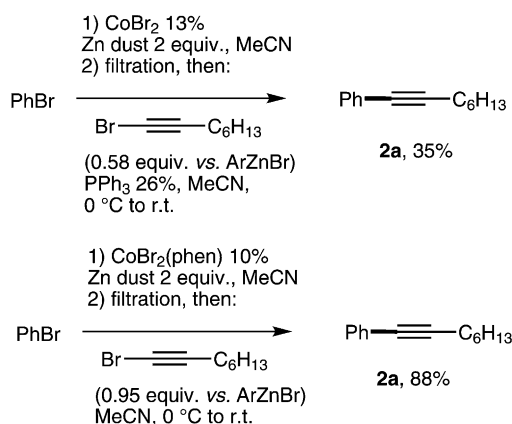
The most important factor influencing the reactivity was the position of the functional group: *para*-substituted arylzinc bromides bearing methoxy (**1a**), acetoxy (**1b**), nitrile (**1d**), ester (**1e**) or trifluoromethyl (**1f**, **1g**) groups afforded the desired product in good to excellent yields; the only unsatisfactory case was the modest yield for the thiomethyl group (**1c**), which presumably reflects poisoning of the cobalt catalyst. When functional groups were introduced at the *ortho* position, the yield decreased sharply (**1h**, **1i**, **1j**). Various functionalities have been also introduced on the aromatic ring of the bromoalkyne, such as methoxy (**1g**, **1k**, **1l**), ester (**1f**) or bromide (**1j**). Finally, the reaction was also possible, with average yield, when a heteroarylzinc bromide was used (**1l**).^[18]



Scheme 1. Scope of the reaction with PPh₃ as ligand.

Next, we sought to expand this methodology to bromoalkynes bearing an alkyl group instead of an aromatic ring. Thus, phenylzinc bromide was coupled with 1-bromooct-1-yne under the optimized reaction conditions using triphenylphosphine as ligand. However, these conditions only gave the desired product in low yield. Thus, we turned to the preformed [CoBr₂(phen)] which allows the formation of the arylzinc species and subsequent cross-coupling reaction in very good yield (Table 1, entry 5; and Scheme 2).^[19]

With this catalyst, the bromoalkyne/arylzinc ratio could be decreased from 1:1.7 to 1:1.05 to avoid arylzincation of the newly formed phenylacetylene. Reaction of *para*-substituted arylzinc bromides with a variety of other bromoalkynes bearing alkyl substituents was remarkably high yielding, and methoxy (**2a**, **2d**, **2g**, **2h**), nitrile (**2f**), chloride (**2g**), acetate (**2i**), thiomethyl (**2j**), as well as phthalimide (**2k**) functional groups were tolerated. This reaction was also possible with a chloroalkyne. Although steric hindrance on the bromoalkyne had almost no influence on the reaction rate or yields (**2j**, **1o**), an *ortho* substituent on the aromatic ring decreased the rate dramatically. Additionally, the presence of the phenanthroline ligand pro-

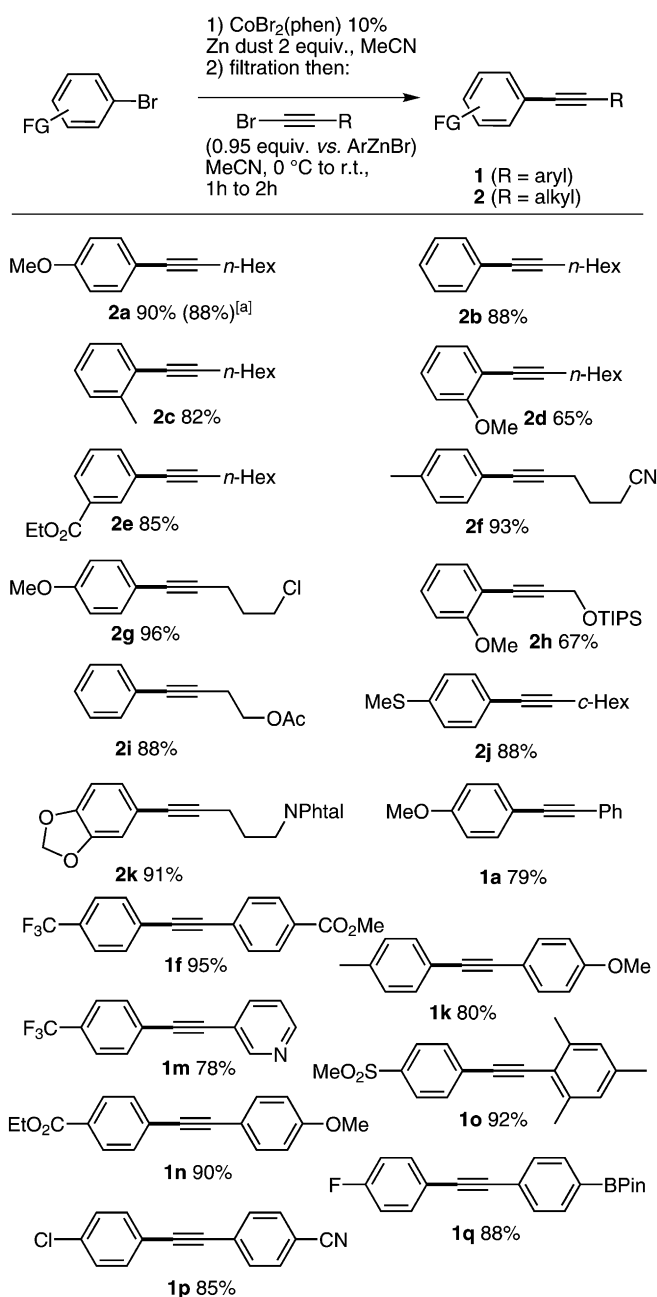


Scheme 2. Arylation of 1-bromooctyne.

tected the catalyst against poisoning in the case of a thiomethyl substituent (**2j**). This new and more convenient cobalt/phenanthroline catalytic system was then applied to the formation of substituted diphenylacetylenes. When electron-donating groups were present on the arylzinc bromide, similar or lower yields were obtained than those when PPh_3 was used as the ligand (compare compounds **1a** and **1k** in Scheme 1 and Scheme 3). However, the yields were largely improved when using electron-poor arylzinc bromides (compare compound **1f** in Scheme 1 and Scheme 3). Moreover, this methodology also has the advantage of limiting waste, as a smaller excess of arylzinc reagent can be used.

When reacted with alkyl-substituted bromoalkynes, electron-withdrawing groups on the arylzinc bromide proved to be problematic: when a mesomeric electron-withdrawing group (e.g., ester, nitrile) was introduced at the *para* position, an intractable mixture of cross-coupling and arylzincation products was observed within 30 min. However, the same substituent could be introduced at the *meta* position without the side reaction occurring (**2e**, Scheme 3). Trifluoromethyl groups were not tolerated in any position (intractable mixtures in the *para* or *meta* position, partial conversion after one week with the *ortho* position). Using the supposedly more reactive iodoalkyne gave the same result as the bromoalkyne (Scheme 4).

To establish the role of the cobalt as a catalyst for this reaction, and to investigate the importance of the solvent, the cross-coupling was first run in a mixture of THF and MeCN using phenylzinc bromide obtained through our cobalt-catalyzed methodology, and then using commercial phenylzinc bromide in the same solvent mixture. In the former case, the cross-coupling was achieved in slightly lower yield than in pure MeCN (63% instead of 70%). No cross-coupling product was detected in the latter case; however, addition of cobalt(II) bromide to the mixture containing the commercial phenylzinc bromide allowed the for-

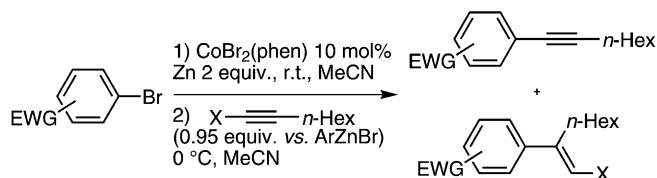


[a] 1-Chlorooctyne was used.

Scheme 3. Scope using $\text{CoBr}_2(\text{phen})$.

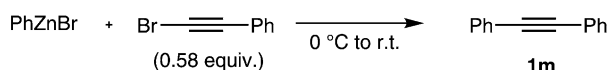
mation of diphenylacetylene in a comparable yield (57%) (Scheme 5) demonstrating its function as a catalyst.

Despite their importance, few cross-coupling methodologies allow the formation of dialkylalkynes. These compounds are usually obtained through nucleophilic substitution, using alkynyllithium or alkynyl-Grignard reagents. However, while recent advances show a better functional group tolerance when using these reagents, they still necessitate carefully controlled reaction conditions, use of additives (TMEDA,



EWG = *p*-CF₃, *m*-CF₃, *p*-CN, *p*-CO₂Et
X = Br, I

Scheme 4. Reaction of electron-poor arylzinc bromides with 1-halo-2-alkylalkynes.



MeCN; PhZnBr prepared with CoBr₂: 70%^[a,b,c]
MeCN:THF (1:1); PhZnBr prepared with CoBr₂: 63%^[a,b,c]
MeCN:THF (1:1); commercial PhZnBr: 0%
MeCN:THF (1:1); commercial PhZnBr, 10% CoBr₂: 57%^[b]

^[a] PhZnBr solution was filtered before use (see above).

^[b] GC yield (vs. bromoalkyne, decane as internal standard).

^[c] PPh₃ (2 equiv. vs. CoBr₂) was added for the cross-coupling.

Scheme 5. Role of cobalt as catalyst.

LiCl), or result in lowered yields.^[20] For these compounds, only few Sonogashira cross-coupling procedures exist, using Pd^[21] or Ni.^[22] Organozinc reagents, on the other hand, can tolerate a much wider range of functional groups, notably esters or nitriles. These reagents can be used in reactions with bromoalkynes, but this transformation requires the use of a stoichiometric amount of copper salts (usually copper cyanide). A single report also described the palladium-catalyzed cross-coupling with bromoalkynes, but the methodology requires cryogenic temperatures (−30 °C).^[14] Satisfyingly, when our cobalt-catalyzed cross-coupling was attempted using CoBr₂(phen), the desired internal alkyne could be obtained, albeit in low yield (Table 2, Scheme 6). A survey of different ligands showed that, while nitrogen-containing ligands were not efficient, *n*-hexylzinc bromide could be coupled with 1-bromooct-1-yne, by using cobalt bromide and 1,2-bis(diphenylphosphino)ethane (dppe) as catalyst. The best solvent for this reaction was pure THF, and an excess of the alkylzinc reagent (2 equivalents) has to be used to insure good yields of internal alkyne. Finally, the reaction proceeds with higher yield at 0 °C.

While exploring the scope of this new reaction, we discovered that, while commercial alkylzinc bromides reacted with high yields, using the reagents prepared with Knochel's method (Zn dust, LiCl, activation with 1,2-dibromoethane and TMSCl)^[23] resulted in significantly lowered yields. When Rieke's finely divided zinc was used,^[24] the results were also disappointing.

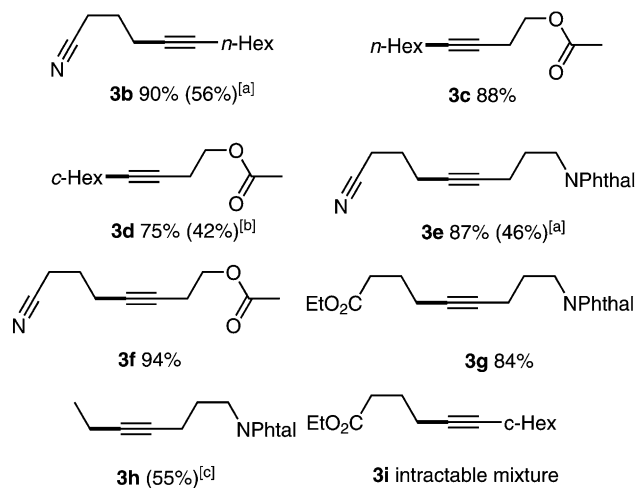
Table 2. Cobalt-catalyzed cross-coupling of alkylzinc bromides with bromoalkynes.

$n\text{-HexZnBr} + \text{Br}-\text{C}\equiv\text{C}-n\text{-Hex} \xrightarrow[\text{THF, r.t.}]{\text{CoBr}_2 \text{ 10\% ligand}} n\text{-Hex}-\text{C}\equiv\text{C}-n\text{-Hex} \quad \mathbf{3a}$		
Entry	Ligand	Yield ^[a]
1	1,10-phenanthroline 10%	30%
2	2,2'-bipyridine 10%	21%
3	quinoline 20%	22%
4	PPh ₃ 20%	72%
5	DPPE 10%	75%
6	DPPE 10% ^[b]	85%
7	DPPE 10% ^[c]	n.d

^[a] GC yield (vs. decane).

^[b] Reaction at 0 °C.

^[c] NMP as cosolvent.



^[a] RZnBr prepared using Knochel's method.

^[b] RZnBr prepared with Rieke zinc containing LiCl.

^[c] RZnX prepared by transmetalation from RMgBr with ZnCl₂.

Scheme 6. Scope of the cross-coupling with alkylzinc reagents. *Reaction conditions:* bromoalkyne (2 mmol), RZnBr (2 equiv.), CoBr₂ (10 mol%), DPPE (10 mol%). RZnBr was prepared using LiCl-free Rieke zinc, and titrated immediately before use.

In these cases, the alkylzinc reagent is supposedly present as an adduct with LiCl. Similarly, alkylzinc species prepared by transmetalation from the corresponding Grignard reagent did not react efficiently. Washing Rieke's zinc with THF allowed us to reduce the amount of LiCl present,^[24b] and allowed the cross-coupling reaction to occur with good to excellent yields, and good functional group tolerance: sensitive functional groups, such as nitriles and esters were well tolerated both on the alkylzinc species and the bromoalkyne. Steric hindrance on the alkylzinc bromide only marginally diminished the yield. However, when

a sterically hindered bromoalkyne was used, the reaction was sluggish (> 24 h) and resulted in an intractable mixture.

To demonstrate the usefulness of this methodology and the interest of cobalt catalysis methods, we sought to combine this arylation reaction with an arylzincation step, which we reported recently.^[17] For that latter reaction, the use of acetonitrile as solvent has been proven to be crucial. Thus, when 1-bromo-4-acetoxybut-1-yne was reacted first with 1.05 equivalents of phenylzinc bromide prepared with $\text{CoBr}_2(\text{phen})$, then with 3 additional equivalents of the same organometallic reagent, in a one-pot procedure, and the resulting vinylzinc quenched with iodine, vinyl iodide **4** was obtained. Negishi cross-coupling with an arylzinc iodide bearing an aminoether side-chain afforded olefin **5**, which is a known intermediate in the synthesis of the selective estrogen receptor modulator Toremifen (Scheme 7).^[25] This modular approach should allow the synthesis of various analogs by replacing the arylzinc halides used for the coupling and arylzincation steps.

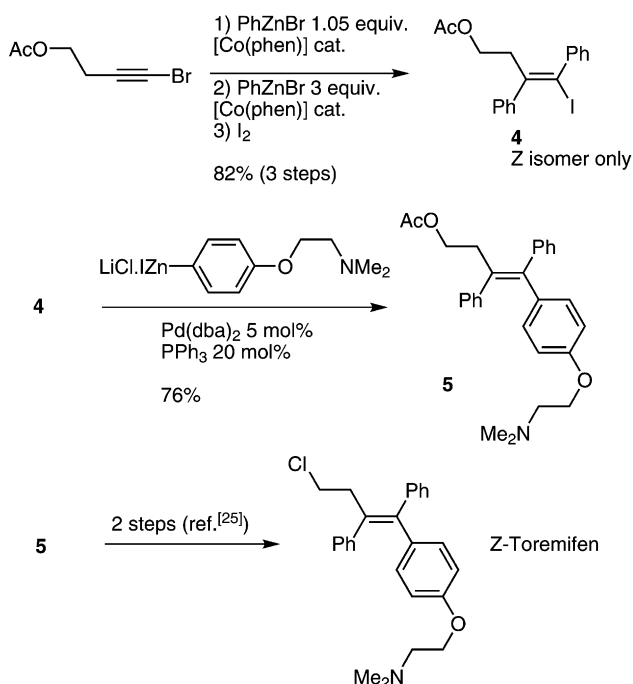
In summary, we have developed a new protocol for the cobalt-catalyzed cross-coupling of arylzinc compounds with haloalkynes allowing the synthesis of various alkynes. This reaction tolerates a large number of functional groups and yields range from modest to excellent. The reaction involving the diarylacetylene from electron-rich arylzinc bromides can be achieved using a cheap and ubiquitous ligand (PPh_3), while in other cases the common ligand 1,10-phenanthroline is preferable. The low price of the catalyst used, and the

mild and “bench-friendly” conditions for the synthesis of the reagents make this reaction an interesting alternative to more classical methodologies. In contrast with the Sonogashira or Heck–Cassar protocols, the steric hindrance of the alkyne partner has almost no influence on the outcome of the reaction. It should also be noted that the alkyne moiety, which is usually the most “precious” one (either expensive or requiring several reaction steps to make), is the limiting reagent in these reactions, and thus does not have to be introduced in excess. In addition, inexpensive aryl bromide can be used as the source of the aryl moiety. We also discovered a highly efficient cobalt-catalyzed cross-coupling reaction allowing the synthesis of dialkylalkynes bearing sensitive functional groups. This new cobalt-catalyzed reaction allows easy access to unsymmetrical internal alkynes, and complements known methodologies. Finally, cobalt catalysis allows an access to Toremifen in a very simple manner.

Experimental Section

General Procedure for the Synthesis and Cross-Coupling of Arylzinc Bromides with Bromoalkynes

To a solution of $\text{CoBr}_2(\text{phen})$ (1 mmol, 399 mg, 10 mol%) and Zn dust (20 mmol, 1.3 g, 2 equiv.) in MeCN (8 mL) were added allyl chloride (0.25 mL, 3 mmol, 30 mol%) and TFA (0.1 mL), at room temperature under vigorous stirring. This caused a rise in temperature, production of gases, and a change of the color of the mixture from blue to orange to dark grey. Once the orange tinge of the mixture had disappeared, ArBr (10 mmol, 1 equiv.) was added. The reaction was followed by GC on iodolyzed aliquots. Once all the starting bromide was consumed, stirring was interrupted; the reaction medium was taken in a 10-mL syringe, and filtered through a syringe filter. The starting flask was rinsed with MeCN (7 mL), which was filtered in the same manner, providing 15 mL of a cobalt-containing arylzinc bromide solution. An internal standard was added (decane or dodecane), and the solution was titrated by GC on an iodolyzed aliquot. The solution was cooled to 0 °C in a water/ice bath, the bromoalkyne (0.95 equiv. vs. ArZnBr) was added, and the reaction was left to warm slowly to room temperature in the bath. The reaction was followed by GC on hydrolyzed aliquots, until all of the bromoalkyne was consumed (1 h 30–2 h). Aqueous hydrochloric acid (ca. 2 M, 20 mL) and Et_2O (20 mL) were then added to the reaction mixture and stirred for 5 min. The phases were separated, and the aqueous phase was extracted 3 times with Et_2O (20 mL). The combined organic phases were washed with water (20 mL) and brine (20 mL), dried over MgSO_4 , and the solvents were evaporated to afford the crude material, which was purified by column chromatography (SiO_2 , petroleum ether/ethyl acetate).



Scheme 7. Formal, stereoselective synthesis of *Z*-Toremifen.

Acknowledgements

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